

BALLA VEEKSHITA NAIDU

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Experience

Senior Software Engineer Software Engineer

May 2025 – Present
May 2024 – May 2025
Hyderabad

Dvara E-Registry — Python, Django, PostgreSQL, AWS

- Engineered and owned **production Python backend services** for a crop-insurance platform, covering **REST APIs, database design, background processing, and deployment**.
- Designed and deployed a **geospatial scoring engine** for multi-temporal satellite datasets, building backend workflows for field-level risk scoring and production API integration.
- Optimized **Django REST APIs** using database indexing and query optimization, profiling and restructuring database access patterns for production endpoints.
- Designed and implemented a **configurable transaction approval engine** in Django, modeling rule-based workflows and integrating them into the transaction processing pipeline.
- Led development of an end-to-end **React + Django farm digitization platform**, implementing GPS-guided data collection, geospatial validation, and offline synchronization; mentored **2 software engineering interns**.
- Engineered a **route optimization system** for field-agent collections using graph-based scheduling, integrating route generation into the collection workflow.
- Designed **event-detection pipelines** integrating Sentinel satellite data and weather APIs to compute drought and flood timelines at **10m geospatial resolution**.
- Developed an **image-based crop damage assessment system** using **YOLOv5, Faster R-CNN, ResNet, and EfficientNet**, achieving **~90% accuracy** on field-collected imagery.

AI/ML Intern

Aug 2023 – Jan 2024
Dharwad

Dvara Solutions — Python, LLMs, Vector Databases, PostgreSQL, ASR

- Built a **real-time voice-enabled RAG application** for querying domain-specific financial documents, integrating **Whisper** for ASR and **Pinecone/FAISS** for semantic vector search.
- Designed a document ingestion and **embedding-management pipeline** using vector databases and PostgreSQL metadata storage, supporting incremental updates and retrieval workflows.
- Implemented **multilingual ASR pipelines** for regional Indian languages using transcription and translation models, enabling cross-language voice interactions.

Technical Skills

Languages: Python, JavaScript

Backend & APIs: Django, Django REST Framework, Node.js, Express, REST API Design, Microservices

Databases: PostgreSQL, MySQL, MongoDB, Database Indexing, Query Optimization

Backend Engineering: Database Design, Background Processing, Caching, Scalability, Concurrency

Cloud & DevOps: AWS, Git, Linux, Deployment

Frontend: React, HTML/CSS

AI/ML: Pandas, TensorFlow, Scikit-learn, YOLOv5, Faster R-CNN, Vector Databases (FAISS/Pinecone)

Achievements

- Solved **700+ algorithmic problems** across LeetCode and GeeksforGeeks.
- Received the **Trailblazer Award** for outstanding contributions and impact at Dvara E-Registry.
- Received the **Top Performer Award** for consistently delivering high-impact engineering outcomes.
- Secured **4th position** in the Inter IIT Case Study Competition.

Projects

MyCV – Resume Builder | *Node.js, Express, MongoDB, React* | [GitHub](#)

- Built and deployed a full-stack resume builder with a **REST API backend**, profile-driven template autofill, PDF export, and persistent user data using MongoDB.

Animal Image Classification | *TensorFlow, Keras, Scikit-learn, NumPy, Pandas*

- Trained and compared VGG-16 (0.93 acc.) and ResNet50 (0.92 acc.) CNN models on **26,000 images across 10 categories**; benchmarked against a gradient boosting baseline (0.85 acc.), demonstrating a 9%+ lift with deep learning.

Education

Indian Institute of Technology, Dharwad (IIT Dh)

2020 – 2024

Bachelor of Technology (B.Tech) in Mechanical Engineering — CGPA: 9.09 / 10